

Ancestors' Gift: Parental Early Exposure to the Environmentally Realistic Pesticide Mixture Drives Offspring Phenotype in a Larger Extent Than Direct Exposure in the Pacific Oyster, *Crassostrea gigas*

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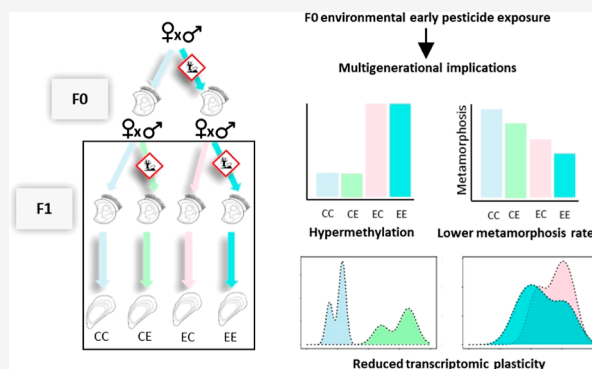
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ABSTRACT: Marine organisms are threatened by the presence of pesticides in coastal waters. Among them, the Pacific oyster is one of the most studied invertebrates in marine ecotoxicology where numerous studies highlighted the multiscale impacts of pesticides. In the past few years, a growing body of literature has reported the epigenetic outcomes of xenobiotics. Because DNA methylation is an epigenetic mark implicated in organism development and is meiotically heritable, it raises the question of the multigenerational implications of xenobiotic-induced epigenetic alterations. Therefore, we performed a multigenerational exposure to an environmentally relevant mixture of 18 pesticides (nominal sum concentration: $2.85 \mu\text{g}\cdot\text{L}^{-1}$) during embryo–larval stages (0–48 hpf) of a second generation (F1) for which parents were already exposed or not in F0. Gene expression, DNA methylation, and physiological end points were assessed throughout the life cycle of individuals. Overall, the multigenerational effect has a greater influence on the phenotype than the exposure itself. Thus, multigenerational phenotypic effects were observed: individuals descending from exposed parents exhibited lower epinephrine-induced metamorphosis and field survival rates. At the molecular level, RNA-seq and Methyl-seq data analyses performed in gastrula embryos and metamorphosis-competent pediveliger (MCP) larvae revealed a clear F0 treatment-dependent discrimination. Some genes implicated into shell secretion and immunity exhibited F1:F0 treatment interaction patterns (e.g., *Calm* and *Myd88*). Those results suggest that low chronic environmental pesticide contamination can alter organisms beyond the individual scale level and have long-term adaptive implications.

KEYWORDS: epigenetics, gene expression, molluscs, DNA methylation, multigenerational, contaminant cocktail



1. INTRODUCTION

The worldwide use of pesticides has led to their presence in all of the Earth compartments.^{1,2} As a consequence, although public authorities try and preserve water quality, continental and coastal water bodies are still contaminated by complex mixtures of pesticides.³

Pesticides threaten all of the integration levels within ecosystems, especially in the marine environment which constitutes their final destination. Indeed, marine organisms are nontarget species known to be impacted by pesticides.⁴ Among them, many nonvertebrate species like molluscs⁵ exhibit external fertilization and their early life stages are exposed to environmental stressors and constitute windows of high susceptibility, potentially leading to long-lasting or delayed effects.^{6,7} The Pacific oyster, *Crassostrea gigas* (Thunberg, 1873; e.g., *Magallana gigas*), is a historically farmed species of great economic interest worldwide. Moreover, it exhibits specific biological characteristics (filter feeder,

sessile, widespread) that explain its long-standing use as a marine ecotoxicological model.^{8–10}

For many years, ecotoxicological studies documented pesticide toxic effects, revealing several outcomes, i.e., embryotoxicity,^{11,12} genotoxicity,^{9,13,14} reprotoxicity,¹⁵ or symbiototoxicity.¹⁶ However, two great challenges are emerging: (i) the study of complex mixtures to better fit environmentally relevant conditions¹⁷ and (ii) the integration of intergenerational and evolutionary perspectives.¹⁸ If a growing body of literature investigates binary and ternary mixtures, the issue of complex cocktails remains less addressed,¹⁹ notably in molluscs where the few existing studies do not allow a satisfactory risk

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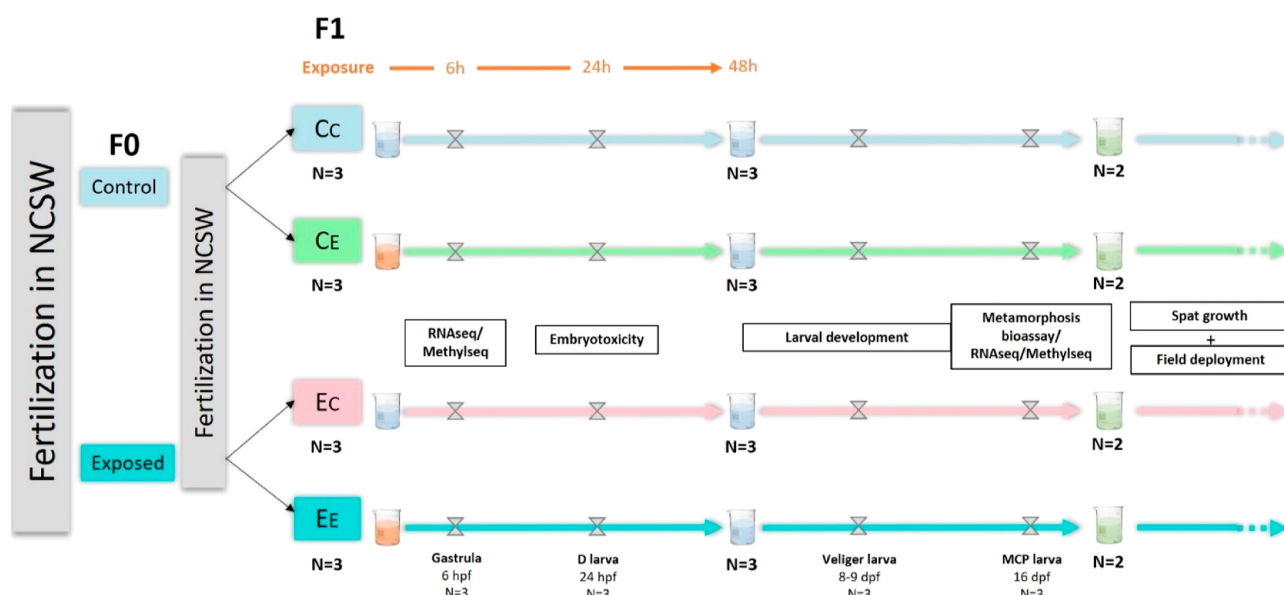


Figure 1. Experimental design. C_C = control–control; C_E = control–exposed; E_C = exposed–control; E_E = exposed–exposed; orange beaker = contaminated seawater; blue beaker = noncontaminated seawater (NCSW, 1 μm filtered, UV-filtered and active coal-filtered, 25 $^{\circ}\text{C}$); green beaker = pretreated seawater (UV-treated, 10 μm filtered); Oo = fertilized oocytes; spm = spermatozoa. N = number of samples collected per tank for each analysis and of replicates per conditions. (The F0 generation was analyzed and previously published in Sol Dourdin et al.⁶ The F1 generation is presented in this paper.)

assessment.^{20–23} On the other hand, intergenerational ecotoxicological studies mostly focus on shorter generation-time species, i.e., zebrafish²⁴ or daphnia,²⁵ which are less ecologically relevant. Considering longer generation-time species like molluscs, the available studies usually investigated the parental effects of exposure during parental gametogenesis,^{26,27} and to our knowledge, the multigenerational effects of early exposures in the Pacific oyster have only been addressed once, in the case of a microbiota study.¹⁰

Because epigenetic marks can persist through generations, epigenetics has become an important tool to investigate the intergenerational issue. In this context, DNA methylation and its link with differential gene expression have gained interest and are increasingly studied in ecotoxicology as a new toxicity end point as well as a tool for transgenerational risk assessment.²⁸ DNA methylation is known to be implicated into gene expression regulation²⁹ and adaptation to environmental changes within and across generations.³⁰ In the oyster, DNA methylation is crucial for the development, notably because it is associated with gene expression.³¹ Exposure of oyster embryos to contaminants such as copper induces development gene methylation and expression defects together with developmental abnormalities.³² In addition, we have recently demonstrated that an early exposure to an environmentally relevant pesticide mixture leads to adverse carry-over effects and the disruption of key developmental stages.⁶ Because DNA methylation is affected by pesticides,³³ the question of the multigenerational impacts and adaptive consequences of an early exposure in the Pacific oyster is asked. This study investigated the impacts of a multigenerational exposure to an environmentally relevant cocktail of 18 pesticides (nominal sum concentration: 2.85 $\mu\text{g}\cdot\text{L}^{-1}$) during the first 48 h after fertilization. A multiscale survey was conducted by combining the assessment of larval development, metamorphosis rate, spat growth, and on-field survival together with transcriptomic, methylomic and genetic approaches.

2. MATERIALS AND METHODS

2.1. Experimental Design. **2.1.1. Chemical Mixture.** The pesticide mixture used was the same as the one used in the F0 generation.⁶ Briefly, the pesticides of the mixture were chosen based on their reported presence in the main French oyster farming areas during the reproduction period of the oyster. The 18 chosen compounds mimic an environmentally relevant pesticide cocktail with a total nominal concentration of 2.85 $\mu\text{g}\cdot\text{L}^{-1}$ (Table S1). Individual stock solutions were prepared from powders in their respective solvents (methanol or ultrapure water) and combined in a mixture stock solution at $\times 100,000$ concentration.

2.1.2. Broodstock: Origin, Conditioning, and Fertilization. The first generation of *C. gigas* oysters (F0) was reared under two experimental conditions during the first 48 h after fertilization: (i) control in noncontaminated seawater (NCSW) or (ii) exposed to the mixture of 18 pesticides (E, 2.85 $\mu\text{g}\cdot\text{L}^{-1}$ nominal concentration). Data and analyses of the F0 generation were already published in Sol Dourdin et al.⁶ One year old individuals from each condition were randomly selected. They underwent broodstock conditioning (NCSW, 8 weeks at 18 $^{\circ}\text{C}$, fed ad libitum with *Skeletonema costatum* and *Isochrysis lutea*) and were then induced to spawn by thermal shock (C, 13 males, 6 females; E, 14 males, 8 females). Oocytes and spermatozoa were collected as the passing fractions on 80 and 20 μm sieves, respectively, and then pooled by sex and condition. A per condition fertilization was then performed as previously described,³² resulting in an E progeny and a C progeny.

2.1.3. Embryonic Development under Chemical Contamination and Larval Rearing. Embryos from each fertilization (C and E parental conditions) were divided into two treatments: (i) control (C, NCSW with solvents [methanol and Milli-Q water representing $5 \times 10^{-5}\%$ and $9.5 \times 10^{-4}\%$ of the total volume], 25 $^{\circ}\text{C}$, 31.1 psu, air bubbling, $n = 3$ tanks, 30 L tanks, 100 embryos $\cdot\text{mL}^{-1}$) and (ii) exposed to the

environmental pesticide mixture (E: $2.85 \mu\text{g}\cdot\text{L}^{-1}$ nominal concentration, 1/100,000 v/v dilution of the mixture stock solution in NCSW, 25°C , $31.1 \mu\text{psu}$, air bubbling, $n = 3$ tanks, 30 L tanks, $100 \text{ embryos}\cdot\text{mL}^{-1}$). This resulted in four experimental conditions: C_C and C_E are embryos from control parents (C parent) that are exposed (C_E) or not exposed (C_C) to the mixture in the present generation. E_C and E_E are embryos from exposed parents (E parent) that are exposed (E_E) or not exposed (E_C) to the mixture in the present generation, respectively (Figure 1). Embryos were maintained under these conditions for 48 h. Thereafter, they were transferred in a flow-through larval rearing system supplied with NCSW (25°C , $33.4 \pm 0.2 \text{ psu}$, 5 L tanks, $5 \text{ L}\cdot\text{h}^{-1}$, $50 \text{ larvae}\cdot\text{mL}^{-1}$, $n = 3$ tanks per condition) with ad libitum equal volumes of *I. lutea* and *Chaetoceros gracilis* from 2 dpf to 6 dpf and then *I. lutea* and *Thalassiosira weissflogii* (homogeneous algae outflow = $1.5 \times 10^6 \text{ mm}^3\cdot\text{L}^{-134}$). At 16 dpf (14 days post exposure), about 50% of the larvae from each condition reached the eye-spotted pediveliger stage (metamorphosis-competent pediveliger [MCP] larva), which reveals the acquisition of the competence for metamorphosis.

2.1.4. Spat Rearing. MCP larvae from each tank were pooled according to their experimental conditions to homogenize larval concentrations. They were then transferred in a flow-through raceway for settlement in NCSW (25.0°C , $32.3 \pm 0.6 \text{ psu}$, $100 \text{ L}\cdot\text{h}^{-1}$) and reared as previously described.⁶ One year old individuals were deployed on-field in La Coupelasse site ($47.026571^\circ \text{ N}/2.030872^\circ \text{ W}$) and checked every 2 weeks for survival and growth based on the methodology described in Fleury et al.³⁵

2.2. Sampling Protocols and Analyses. **2.2.1. Seawater Chemical Analyses.** Seawater chemical analyses were performed for organic and metallic compounds at T0 (before embryo incubation) following the protocols described in ref 6. Metallic compounds were measured using ICP–MS (I-CAP-TQ, Thermo), and organic compounds were analyzed using liquid chromatography–tandem mass spectrometry (LC–MS/MS) as described in Sol Dourdin et al.⁶

2.2.2. Ecotoxicology Tests and Morphometry. The embryotoxicity test was conducted in triplicate according to the previously described protocol,⁶ following the standardized embryo–larval bioassay ISO 17244:2015 based on the counting of larval abnormalities. During larval rearing, larvae were sampled in each flow-through tank every 2–3 days and stored in a 0.1% seawater–formaldehyde solution until image analysis as described in ref 6. The metamorphosis bioassay was conducted on competent eye-spotted pediveliger larvae retrieved on a $150 \mu\text{m}$ sieve (14 dpf, 12 days post exposure) in 12-well microplates using epinephrine (Sigma-Aldrich) at a 10^{-4} M final concentration as described in ref 6. Spat growth and on-field survival monitoring were conducted following the protocols used for the F0 generation.⁶

2.2.3. Statistical Analyses for Ecotoxicology and Morphometry Data. Results are presented as the mean $\pm$ 95% confidence interval (CI) of independent replicates. All the data were processed and analyzed using R/Bioconductor.³⁶ Multiple comparisons were carried out using two-way ANOVA ($\sim\text{F0_Treatment} \times \text{F1_Treatment}$) to test for interactions, followed by Newman–Keuls' post hoc test. The normality (Shapiro's test) and homoscedasticity (Bartlett's test) of residues were systematically checked using the performance R package (performance_0.10.1).³⁷ The Kruskal–Wallis's (KW) test was performed as nonparametric alternative to

compare between the four F1 conditions, and the Wilcoxon's test was used as nonparametric alternative to compare between the two lineages. Survival rates were represented as Kaplan–Meier curves and tested for significant differences using a log-rank test with the survival R package (survival_3.3-1).³⁸

2.2.4. Molecular Analyses. **2.2.4.1. Sampling Protocol, RNA and DNA Extractions, and Sequencing.** Individuals were sampled at gastrula (6 hpf, 6 h of exposure, 50,000 embryos per sample, 1 replicate per 30 L tanks) and MCP larva stages (16 dpf, 14 days post exposure, 1500 animals per 5 L tank). RNA and DNA extractions were performed using Trizol (Invitrogen) and E.Z.N.A Mollusc DNA Kit (Omega Bio-tek, Norcross, USA), respectively, as previously described.⁶ RNA sequencing (RNA-seq, paired-end $2 \times 150 \text{ bp}$, Illumina NovaSeq 6000) and DNA methylation sequencing (Methyl-seq, paired-end $2 \times 150 \text{ bp}$, Illumina NovaSeq 6000) were performed at Genome Quebec, Montreal, Canada. RNA sequencing led to ca. 1.8 billion of paired-end reads, i.e., 77 ± 0.12 million reads per sample, with an average quality score of 36 ± 0 and a presumptive ca. $100\times$ transcriptome coverage. Genomic DNA sequencing led to ca. 1.9 billion of paired-end reads, i.e., 79 ± 0.15 million reads per sample, with an average quality score of 35 ± 0 and a presumptive ca. $35\times$ coverage. Bioinformatic data processing was carried out on Ifremer's high-performance computing cluster (Datarmor).

2.2.4.2. RNA-seq Analyses. Raw reads were checked using FastQC (fastqc_v0.11.9):³⁹ all bases were above 30Q_c score. Reads were mapped to the indexed reference genome GCA902806645v1⁴⁰ using STAR (star_v2.7.9a)⁴¹ with the following parameters: —alignIntronMin 20—alignIntronMax 1,000,000. The resulting mapping rate was about $48.9 \pm 0.1\%$. Gene expression was quantified by counting the number of reads mapped on mRNA sequences, and only genes exhibiting 10 reads or more were kept.

Principal component analyses (PCAs) were performed with the “plotPCA” function from DESeq2 considering the 1000 more variable genes. The differential expression analyses were performed with the DESeq2 R package⁴² after *vst* normalization. The Wald's test was used to check for parental effect ($\log_2$ fold change > 0.5 , α threshold = 0.05) and the likelihood-ratio test (LRT) to check interaction between F0 and F1 treatments (α threshold = 0.1). Heatmaps of the differentially expressed genes (DEGs) were built from the normalized count matrix with correlation distance using the pheatmap R package (pheatmap_1.0.12).⁴³ Rank-based gene ontology (GO) term enrichment analysis was performed with the GO_MWU method using $\log_2$ fold change (L2FC) values.⁴⁴

The global transcriptomic shift upon pesticide exposure was analyzed using a discriminant analysis of principal components (DAPCs) on normalized counts.⁴⁵ The DAPC was performed on a per-lineage (F0 treatment: control vs exposed) basis using the adegenet R package (adegenet_2.1.5).⁴⁶ This analysis allows the description of a population using variables maximizing intergroup divergence and minimizing intragroup divergence (namely, the discriminant function). The same function was used to discriminate samples within each lineage in the two F1 treatment groups: control and exposed. Coordinate distribution of each F0–F1 treatment groups along the first DAPC was fitted, and their difference was tested against a generalized linear mixed model using the Markov chain Monte Carlo (MCMC_{glimm}) method ($\text{LD1} \sim \text{F0Treatment}$).

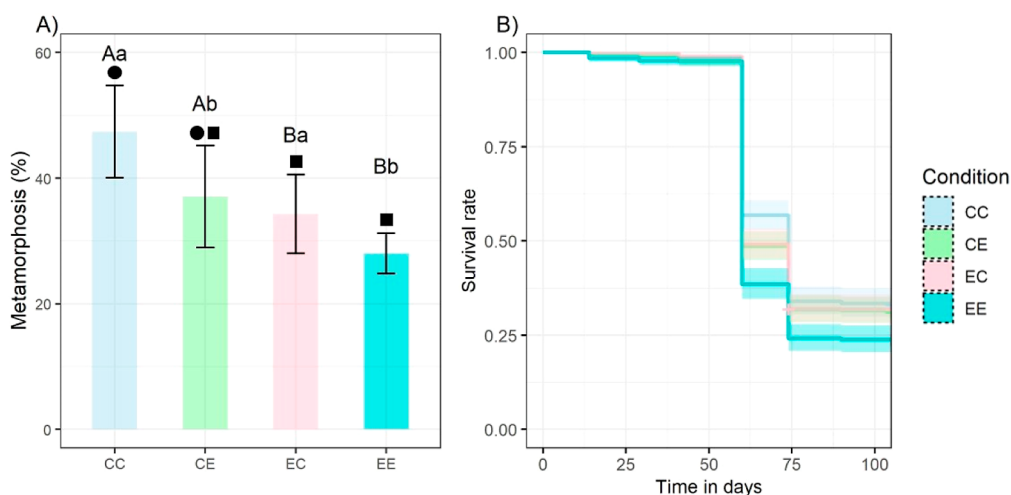


Figure 2. (A) Epinephrine-induced metamorphosis rate (%). Letters refer to statistically significant differences between parental treatment (upper-case) and F1 treatment (lower-case). Two-way ANOVA, $p < 0.001$. Black shapes refer to statistical differences between the four F1 conditions. One-way ANOVA, $p < 0.05$. Error interval represents the 95% CI. (B) Survival rate of spats in the field. Error interval represents the 95% CI (log-rank test, $p < 0.01$).

ment + F0Treatment:F1Treatment) implemented in the MCMCglmm R package (MCMCglmm_2.33).⁴⁷

2.2.4.3. Methyl-seq Analyses. Raw read quality was checked with FastQC (fastqc_v0.11.9).³⁹ fastp (fastp_v0.20.1)⁴⁸ was used to trim Illumina adapters, the last 20 poor quality bases of reads 1 and 2, and filter reads by length ($125 \geq \text{bp}$). Trimmed reads were mapped to the indexed GCA902806645v1 reference genome⁴⁰ with BWA-Meth (bwa-meth_v0.6.1),⁴⁹ leading to a $52.9 \pm 0.1\%$ mapping rate. Methylation bias and methylation calling were performed using MethylDackel (methylDackel_V0.6.1) (<https://github.com/dpryan79/MethylDackel>) with the following parameters: —minDepth 10—OT 5,125,5,125—OB 5,125,5,125.

Methylation data were summarized on a per-tile basis (tile = 500 bp) using the methylKit R package (methylKit_1.16.1).⁵⁰ PCA was performed with the “PCA()” function from the FactoMineR R package (FactoMineR_2.4) (1000 more variable regions),⁵¹ Wald’s test was used to check the parental effect ($|\Delta| = 10\%$, α threshold = 0.05), and the LRT was performed to check the interaction between F0 and F1 treatments (α threshold = 0.1). Differentially methylated region (DMR) coordinates were intersected with gene coordinates from the GCA902806645v1 genome assembly, using Bedtools intersect (bedtools_v2.30.0),⁵² to identify DMR-related genes (DMGs). The differential methylation of DMGs was estimated as the mean of the differential methylation of the related DMRs. Rank-based GO term enrichment analysis was performed with the GO_MWU method using Fisher’s exact test.⁴⁴

The relationship between gene expression and DNA methylation was explored based on the 5000 more variable genes and regions using the Projection to Latent Structure (PLS) method from the mixOmics R package (mixOmics_6.14.1).⁵³

2.2.4.4. Genetic Divergence. Biallelic single-nucleotide polymorphisms (SNPs) were identified from the RNA-seq data of the F0 and F1 generations using the nf-core rnavar pipeline 1.0.0.⁵⁴ Variant callings were merged together and then filtered using BCFtools v1.17⁵⁵ with the following parameters “-g 10-i ‘TYPE = ‘snp’ & N_ALT = 1 & min(FORMAT/DP) ≥ 4 & max(FORMAT/DP) ≤ 2000 &

min(QUAL) ≥ 100 & min(MQ) ≥ 30 & min(AC) ≥ 3 & F_MISSING < 0.1 ”. The allele frequencies were computed for each SNP and each sample, and they were plotted on a PCA. In order to estimate the genetic divergence between both lineages, the absolute allele frequency difference (AFD) was used as alternative to the F_{ST} metrics,⁵⁶ using a script developed in Andres et al.⁵⁷ Whether and to what extent SNPs may account for methylation differences (i.e., the physical link between SNPs and methylation variations) were assessed by overlapping SNP coordinates with between-lineage DMR coordinates using Bedtools Intersect (bedtools_v2.30.0).⁵²

3. RESULTS

3.1. Chemical Contamination. The measured concentrations (MCs) in the control tanks were under the detection or quantification limits for all of the organic contaminants (Table S2). In the exposed batches, the MC of metallic contaminants was consistent with the nominal concentrations. Regarding organic contaminants, the MCs are consistent with the nominal concentrations excepted for atrazine, diuron, carbendazim, acetochlor, and simazine that exhibit concentrations approximately 2-fold lower than expected.

3.2. Embryotoxicity Test, Larval Development, and Metamorphosis Bioassay. The embryotoxicity test conducted at 24 hpf did not reveal major morphological alterations, and there was no difference between the four conditions (C_C : $92.9 \pm 1.9\%$; C_E : $90.7 \pm 2.7\%$; E_C : $90.1 \pm 2.6\%$; E_E : $89.8 \pm 3.6\%$; two-way ANOVA, $p > 0.05$). The larval sizes were different at days 2 and 13 (Figure S1). At day 2, the C_C larvae were larger than the C_E , E_E , and E_C conditions (KW test, $p < 0.05$), and at day 13, C_E was greater than E_E (KW test, $p < 0.05$). At 16 days post fertilization, the larvae from all conditions reached the competence for metamorphosis, with no statistic difference in eye-spotted pediveliger larvae percentage (C_C : 38.7 ± 29.8 ; C_E : 64.0 ± 30.1 ; E_C : 41.7 ± 19.3 ; E_E : 37.7 ± 34.8 ; two-way ANOVA, $p > 0.05$).

The metamorphosis rate was higher in the control lineage than in the exposed lineage, and there was a significant effect of the F1 exposure, which induced a lower metamorphosis rate

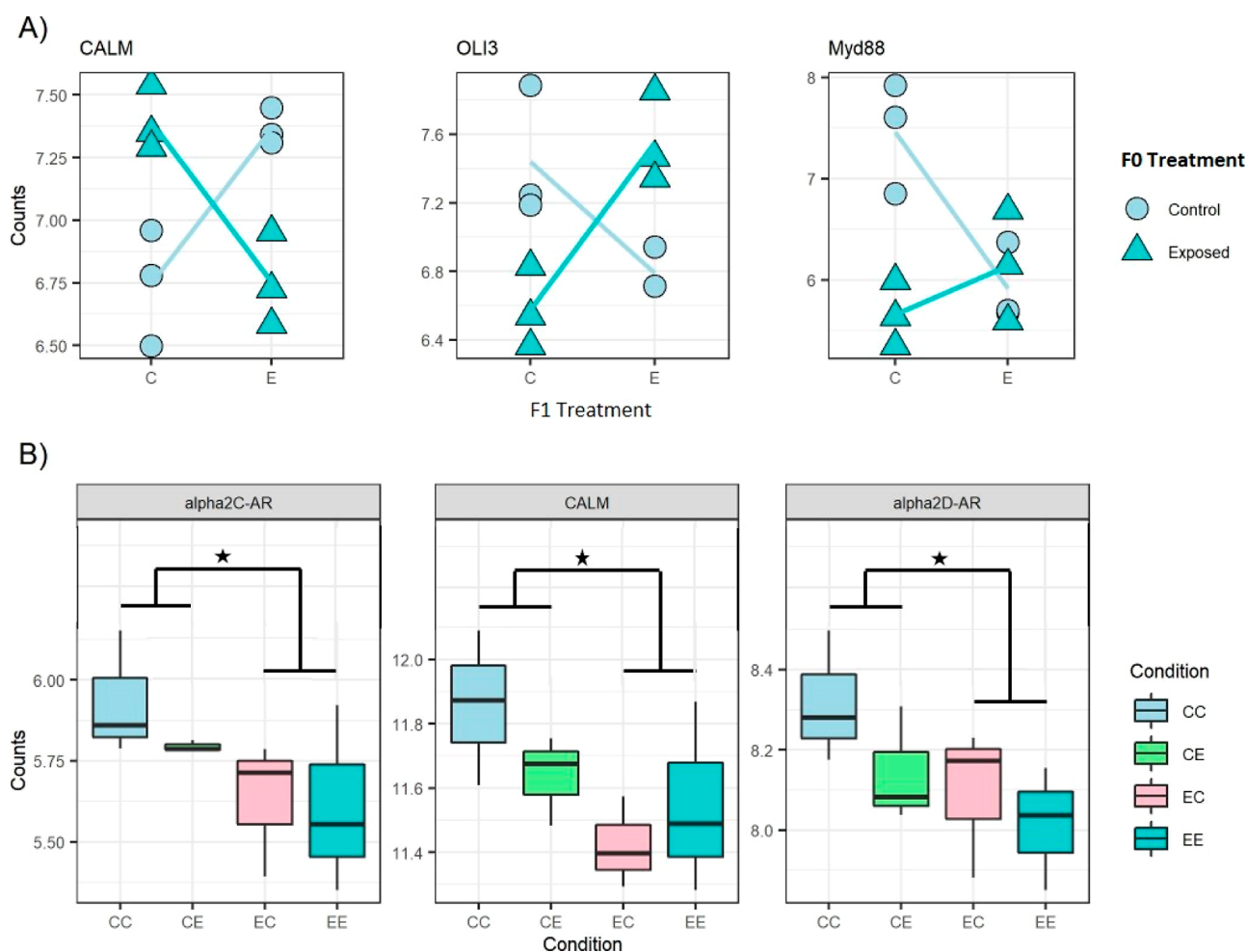


Figure 3. (A) RNA-seq normalized count plot of three genes exhibiting the significant F1:F0 interaction pattern of expression in gastrula embryos, LRT, $p \leq 0.1$. (B) Expression level of metamorphosis-related genes based on RNA-seq normalized counts. One-tailed Wilcoxon's test between lineages, *: $p \leq 0.05$. C = control treatment; E = exposed treatment.

(Figure 2A). No interaction was observed between F0_Treatment and F1_Treatment (two-way ANOVA, $p > 0.05$).

3.3. Spat Growth and On-Field Survival. Spat total weight was statistically different between conditions from months 1 to 3, from months 5 to 8, and at month 11 post fertilization; however, no clear trend was observed (Figure S12). E_E group placed in the field had a lower survival rate than C_C, C_E, and E_C groups (24, 33, 32, and 32%, respectively; log-rank test, $p < 0.005$, Figure 2B).

3.4. Gene Expression. In gastrula embryos (6 h pesticide exposure), samples were discriminated by the F0 treatment along the second PCA component that explained 14% of the variance (Figure S3A). There were 451 DEGs between parental treatments (Table S3, Wald's test, $p < 0.05$, |L2FC| threshold = 0.5). Of those genes, 203 were induced and 248 were repressed in the exposed lineage compared to the control (Figure S4A). The GO term enrichment analysis revealed that several biological processes such as mitochondrial respiratory or peptide biosynthetic processes were repressed (Figure S5A). Instead, biological processes implicated in nervous system development, stress response, and signaling pathways were induced in the offspring from exposed parents. There were 20 genes exhibiting F1:F0 treatment interaction pattern of expression (LRT, $p < 0.1$) (Figure S6A and Table S4). Three patterns of interaction were observed (Figure S6C): (i)

the control lineage expression increases in response to pesticide exposure, while the exposed lineage expression decreases, e.g., calmodulin (CALM, Figure 3A); (ii) the control lineage expression decreases, while the exposed lineage expression remains constant, e.g., myeloid differentiation primary response protein 88 (Myd88, Figure 3A), and (iii) the control lineage expression decreases, while the exposed lineage expression increases, e.g., oligodendrocyte transcription factor 3 (OLI3, Figure 3A).

In MCP larvae, transcriptome-wide signatures related to the parental exposure were no longer detected with the global PCA (Figure S3B). Anyway, the differential analysis based on the F0 treatment revealed 241 DEGs (Table S3, Wald's test, $p < 0.05$, |L2FC| threshold = 0.5), among which 130 were induced and 111 were repressed in the exposed lineage compared to the control (Figure S4B). The GO term enrichment analysis showed the repression of biological processes linked to nervous system development and regulation and signal transduction, while lipid metabolism, translation, or cell motility was induced in the exposed F0 offspring (Figure S5B). Five genes exhibited the F1:F0 treatment interaction pattern according to the pattern (i) only (Figure S6B and Table S4). Among those genes, three are related to stress response: HSP68, HSP70, and stress-induced protein (Table S4).

Three genes putatively implicated in the response of MCP larvae to epinephrine ($\alpha 2C$ -AR, $\alpha 2D$ -AR, and CALM) (Figure 3B) were individually tested for differential expression between lineages. All of them exhibited differential expression (one-tailed Wilcoxon's test, $p \leq 0.05$), being more expressed in the control than in the exposed lineage.

The pesticide-induced transcriptomic plasticity was estimated from the shift in sample distribution on the first DPCA component built on the normalized count matrix (Figure 4).

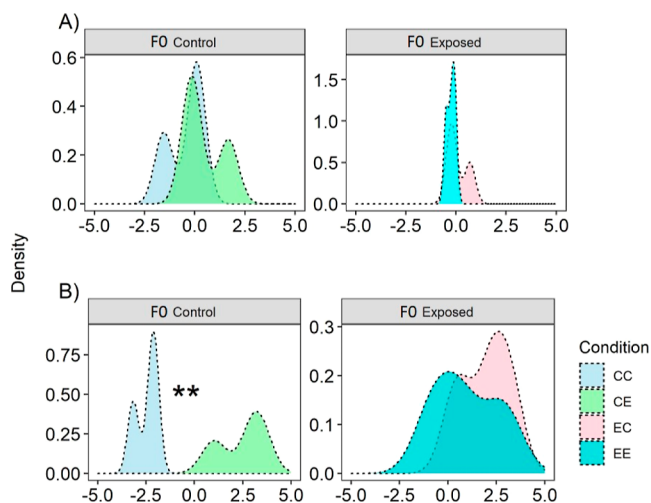


Figure 4. Transcriptomic plasticity: coordinate distribution of each condition along the first DAPC for gastrula embryos (A) and metamorphosis larvae (B). Samples were discriminated against the same function (**: MCMCglmm p -value < 0.01).

The expression shift was not significant for the parental groups in gastrula embryos (MCMCglmm, $p > 0.05$). In MCP larvae, the expression shift was significant in the control lineage but not in the exposed lineage (MCMCglmm, $p = 0.006$ and $p > 0.05$, respectively) (Figure 4), indicating that transcriptomic response to pesticide exposure is attenuated in MCP larvae from parents that experienced the same stress.

3.5. Differential Methylation Analysis. In gastrula embryos, overall methylation was strongly influenced by the F0 treatment. Samples were discriminated by lineage along the first PCA component, counting for 89.75% of the variance (Figure S7A). Differences in methylation preferentially affected intergenic regions, which exhibit 15% of the constitutive

methylation but 27% of the DMRs observed in gastrulae (χ^2 test, $p < 0.05$, Figure S8). There were 4349 DMRs (hypomethylated: 435, hypermethylated: 3914; Wald's test, $p < 0.05$, |Difference| threshold = 10%) between exposed and control lineages, intersecting 2345 genes (Table S5). Of those genes, 89.5% were hypermethylated. Enriched biological processes are mainly related to cell fate, cell junction, and muscle development (Figure S9A). Overall, 84 DMR-related genes exhibited F1:F0 treatment interaction patterns (Figure S10A, Table S6). In MCP larvae, overall methylation did not discriminate on PCA for neither F0 nor F1 treatment (Figure S7B). In MCP larvae, there were 550 DMRs (hypomethylated: 178, hypermethylated: 372; Wald's test, $p < 0.05$, |Difference| threshold = 10%) between both lineages, intersecting 337 genes of which 68.8% were hypermethylated. Regarding GO term analyses, enriched biological processes are mainly related to development and organismal-level homeostasis (Figure S9B). There were 69 DMR-related genes exhibiting F1:F0 treatment interaction patterns (Figure S10B, Table S6). In gastrula embryos, 51 genes were differentially expressed and methylated (Table S7), but there was no correlation between the level of differential methylation and the differential expression for those genes (Pearson's correlation test, $p > 0.05$). None of the differentially methylated genes were differentially expressed in MCP larvae. The PLS analysis revealed that DNA methylation was the main driver discriminating samples at the gastrula stage (Figure S11). For methylation, the first latent component accounted for 66.1% of the variance and discriminated C_C and E_E conditions. For the gene expression, the second axis explained 56% of the variance without discriminating samples. In the MCP larva stage, the PLS analysis did not allow us to discriminate conditions (Figure S12).

3.6. Genetic Divergence. The projection of allele frequencies on a PCA (Figure 5) did not reveal genetic divergence in the F0 generation between exposed and control individuals. A clear differentiation was observed between parents (F0) and offspring (F1) which discriminate along the first component (17.56% variance). In the F1 generation, individuals from each pedigree also discriminate along the second axis of the PCA (11.28% variance). The AFD calculation resulted in a similar differentiation between both F1 lineages (AFD = 0.079) and between F0 and offspring from the control pedigree (AFD = 0.077) or the exposed pedigree (AFD = 0.083). The intersection of DMRs and SNPs

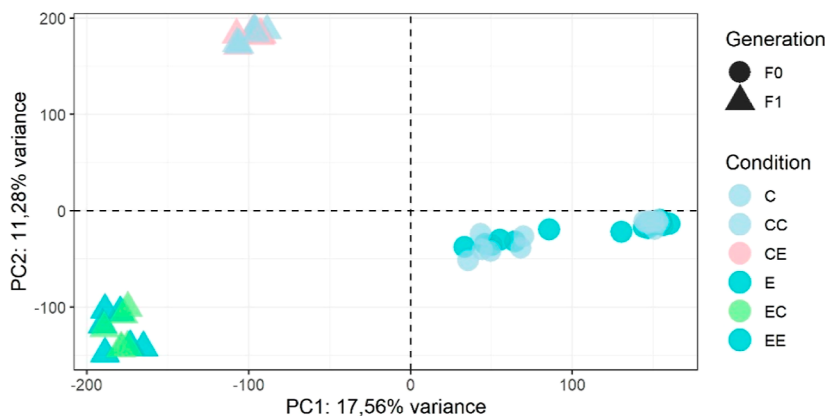


Figure 5. PCA of the allele frequencies related to the SNPs existing between individuals from F0 and F1 generations.

coordinates revealed that the overlapping concerns 0.8% of SNPs and 6% of DMRs in gastrula embryos and less than 0.0001% of SNPs and 0% of DMRs in the MCP larva.

4. DISCUSSION

This study investigated two of the main challenges in ecotoxicology: the complex chemical mixtures¹⁹ and the intergenerational effects.⁵⁸ Three questions were addressed: (i) does an early exposure to environmentally relevant pesticide mixture induce multigenerational effects in *C. gigas*? if so, (ii) does multigenerational inheritance trigger adaptive mechanisms? and (iii) does DNA methylation contribute to the transmission of environmentally induced effects across generations? To answer, we assessed molecular and phenotypic end points in a multigenerational exposure experiment.

Organic contaminants were under detection or quantification limits in all of the unexposed batches (C_C and E_C). Besides, the mean copper and cadmium concentrations in control conditions were representative of the recently MCs in the Loire estuary.⁵⁹ Those results confirmed the efficiency of the active carbon treatment and the absence of metallic contamination in the experimental system. Moreover, the MCs in seawater from all exposed batches (C_E and E_E) were mostly consistent with expected nominal concentrations, except for atrazine, diuron, carbendazim, acetochlor, and simazine that were about 2-fold lower than expected (Table S2). However, the total MC in contaminated batches ($2.75 \pm 0.063 \mu\text{g}\cdot\text{L}^{-1}$) was lower but within the same range than for the F0 generation ($3.75 \pm 0.014 \mu\text{g}\cdot\text{L}^{-1}$).⁶

We chose to investigate gene expression and DNA methylation in the gastrula and MCP larva stages because they correspond to important time windows of direct (gastrula) or delayed (MCP larva) sensitivity to pesticide exposure, where impairments could lead to measurable functional consequences. However, gene expression and DNA methylation are highly dynamic during development,³¹ and it is not excluded that investigating those life stages fails to thoroughly encompass all the possible consequences of pesticide exposure. Therefore, our results should be considered with appropriate caution regarding this point.

Parental effects were observed at both the DNA methylation and the gene expression levels. A clear hypermethylation trend was observed in gastrula embryos as well as MCP larvae from exposed parents (Table S5). Regarding gene expression, gastrulas were clearly discriminated along the second axis (14% variance) based on their origin rather than on the ongoing pollutant stress (Figure S3A), while the first axis (68% variance) is likely to be driven by the sampling process. Overall, 451 genes were differentially expressed at that stage. In gastrulas, offspring from exposed parents exhibited a repression of genes involved in cellular metabolism, while genes related to nervous system development and locomotion were induced (Figure S5A). In MCP larvae, the parental effect still drove the expression of 241 genes, with the repression of biological processes linked to the nervous system structure and development in MCP larvae from exposed parents (Figure S15B). Interestingly, a recent study revealed an upregulation of genes involved in the nervous system remodeling just before the initiation of metamorphosis in *C. gigas*.⁶⁰ The results of the epinephrine-induced metamorphosis test indicate a disruption of the metamorphosis process in the exposed F0 offspring. Indeed, the exposed F1 larvae display lower metamorphosis rates compared to their control counterparts, a trend that was

already observed in the parental (F0) generation.⁶ Such phenotypic impairment is consistent with the repression of CALM and two adrenergic-type receptors (AR) genes in larvae from exposed parents, which are key factors of the epinephrine-response pathway in oysters.⁶⁰ These delayed effects on metamorphosis and thereby recruitment highlight the multigenerational consequences of an early exposure of *C. gigas* embryos to pesticides.

However, the assessment of parental effects being the result of nongenetic inheritance of chemical-induced effects requires to detangle nongenetic from genetic effects, which is recognized as a highly tricky task.⁶¹ Indeed, epigenetic mechanisms could mediate nongenetic effects,⁶² and epigenetic divergence could be faster and predominant over genetic divergence in rapid adaptation of oysters.⁶³ Besides, the data herein do not allow to quantify the relative prevalence of genetic and epigenetic effects. In addition, the Pacific oyster has a highly variable genome, and laboratory experiments are likely to increase genetic drift⁶⁴ or selection,⁶⁵ which may explain the genetic divergence observed here between control and exposed lineages. The interlineage divergence in the F1 generation is also very similar to the divergence observed between parents and offspring and may, therefore, be considered as a consequence of the reproduction experiment design. Nevertheless, in the present work, the hypothesis of multigenerational effects *per se* may be supported by results at the phenotypic and molecular scales. First, the similar metamorphosis rates between the C_E , E_C and E_E groups (i.e., exposed larvae from control parents and both control and exposed larvae from exposed parents) reinforce the idea that this end point is not the consequence of a particular genotype resulting from the protocol of reproduction (Figure 2A). The same conclusion can be drawn from the transcriptomic plasticity analysis of the MCP larvae. The gene expression profiles of the C_E , E_C , and E_E conditions were similar (Figure 4), suggesting that expression patterns are likely to be induced by the early exposure rather than the result of a genetic divergence. Moreover, the limited relationship between SNPs and methylation differences at both gastrula and MCP larva stages reinforces the idea that methylation differences are independent of the (limited) genetic variation between pedigrees. We, therefore, conclude that an early exposure to an environmentally relevant pesticide mixture can have multigenerational effects at multiple biological levels in the Pacific oyster, including the alteration of critical developmental steps like metamorphosis. This result supports a better consideration of the inheritance of toxicity in future studies.⁵⁸

Repeating the chemical stress in the F1 generation allowed us to investigate whether multigenerational inheritance can trigger adaptive mechanisms. F1:F0 interaction patterns were observed in gastrula and MCP larvae at the molecular level, suggesting an adaptive phenomenon. Some interactions are likely to benefit individuals from their exposed parents. For instance, in gastrula embryos, the Myd88 protein, implicated in the innate immune response via the Toll/interleukin-1 receptor (TIR) family signaling pathway,^{66,67} was repressed in the control lineage under pesticide exposure and remained constant in the exposed one (Figure 3B). Later on, the induction of stress-related proteins observed in MCP larvae from control parents after pesticide exposure was mitigated in those from exposed parents. However, nonbenefit interactions were also observed. For example, Calmodulin (Calm) is involved in the cellular calcium homeostasis and biomineral-

ization process in marine invertebrates,⁶⁸ and it is commonly upregulated in oysters facing ocean acidification.^{69,70} Here, in the case of multigenerational exposure to pesticides, the expression of *Calm* is repressed in MCP larvae from the exposed progeny, while it is induced in the control progeny. Therefore, this experiment revealed the possibility for multigenerational exposure to increase the potential negative impact of chemicals on genes involved in shell formation in mollusc larvae, which has already been highlighted in several studies.^{6,32,71} Nonetheless, because the gene expression is highly dynamic during the oyster development, the interaction patterns could also be interpreted as a consequence of the subtle effect of the pesticides on the developmental dynamics, which was suggested by the significant differences in the larval size observed at days 2 and 13 post fertilization. In addition, our observation that the influence of the chemical exposure on the epinephrine-induced metamorphosis rate in MCP larvae was the same in both progenies does not support the presence of strong interaction mechanisms at higher biological levels. However, the field survival rate presented a negative interaction pattern because a higher mortality was associated with the early exposure in spat descending from exposed parents, while no significant effect was measured in the control progeny being exposed. Overall, those results are in line with several studies in marine invertebrates documenting the multigenerational effects of environmental modifications,⁷² which do not report a clear orientation of the influence of the parental exposure toward profitable or detrimental modifications in the offspring. Investigation including more generations could help decipher the adaptive value of these modifications.

Another hypothesis of this study was the role of DNA methylation in mediating environmentally induced effects, which is now commonly admitted.^{73,74} First, in gastrula embryos, DMRs were significantly enriched in intergenic regions (Figure SI9), as already observed in F0,⁶ although dedicated studies are needed to clarify functional consequences of the methylation of noncoding regions and transposons. Second, it is remarkable that pesticide-induced differential methylation followed opposite trends in the F0 (hypomethylation) and F1 (hypermethylation) generations. Such “rebound” has already been observed in oysters after early microbial exposure,¹⁰ but it remains unexplained. Differential DNA methylation revealed F1:F0 treatment interaction patterns, confirming its susceptibility to environmental stressors,⁷⁵ and the PLS analysis indicated that the overall DNA methylation was the main factor discriminating the F1 conditions (i.e., gastrula, first latent component: 66.1% variance). However, there was no direct correlation between differential methylation and gene expression (only 51 DMG and DEGs in gastrula among 2345 DMGs). Such limited or absent correlations have also been observed in fish models,^{76,77} in Anthozoa and Hexapoda,⁷⁸ and in oysters^{10,26,79,80} and can be interpreted in different ways. First, DNA methylation is part of a wide and complex epigenetic network of gene expression regulation mechanisms.^{29,81} Besides, the modification of the genome-wide gene expression under stress condition (Figure 4) was already observed in corals in response to changing habitat⁴⁵ or in bivalve species in response to temperature stress.^{82,83} In our study, the reduction of transcriptomic plasticity after pesticide exposure in MCP larvae from exposed parents was associated with a global hypermethylation trend. In this context, the implications of DNA methylation changes can be considered at a higher-order organization level. If in our

study specific gene body differential methylation does not trigger differential expression, a genome-wide hyper- or hypomethylation could substantially modify the chromatin structure⁸⁴ and global transcriptional potentialities.^{85,86} This could be mediated by potentially DNA methylation-dependent features like transcriptional bursts^{87,88} or alternative splicing.^{89,90} However, alternative splicing was found to be a relatively scarce mechanism potentially occurring in approximately 16% of oyster genes⁹¹ (against 60% in *Drosophila melanogaster*⁹²). Besides, a brief analysis of the alternative splicing events existing between control and exposed lineages failed to highlight the evident relation with the observed differential methylation (the analysis is presented in the Supporting Information). Understanding the mechanisms responsible for the functional relationships between methylation changes and the complex process of gene expression, therefore, requires further studies.

To our knowledge, this work is the first investigation of the multigenerational implications of early exposure to an environmentally relevant pesticide mixture in the Pacific oyster. Our results revealed that an early chemical exposure can substantially affect individuals at critical developmental stages and has multigenerational consequences. On one hand, we observed the F0 to F1 transmissions of environmentally induced defects (i.e., lower epinephrine-induced metamorphosis rate) and, on the other hand, transcriptomic and methylomic data at gastrula and MCP larva stages suggested potential adaptive or maladaptive phenomenon. This suggests that low, chronic environmental contamination can have substantial adaptive consequences on nontarget species. These considerations are of great interest for ecotoxicology. They highlight the need for long lasting and multigenerational studies on environmentally relevant concentrations of contaminants and the need to improve our understanding of how epigenetic mechanisms act within gene regulation networks to efficiently integrate those new “epimarkers” in ecotoxicology frameworks.

■ ASSOCIATED CONTENT

Data Availability Statement

RNA-seq and Methyl-seq data are available at the European Nucleotide Archive under the accession numbers: PRJEB58194 and PRJEB58545, respectively, and metadata are presented in the Supporting Information. All nonmolecular data are available in the Supporting Information. Codes used for bioinformatics and statistical analyses are available at <https://gitlab.ifremer.fr/ccem-public/PESTO-project>.

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.est.3c08201>.

Larval length, spat total weight, sample PCA projection based on the gene expression counts, normalized counts of DEGs, enriched biological processes in gastrula and MCP larva, normalized counts of the F1:F0 treatment, sample PCA projection based on the tile methylation level, genomic features, enriched biological processes in DMGs, methylation level of DMR-related genes, and projection to latent structure analysis in gastrula embryos and MCP larvae; and the analysis of the alternative splicing and its relation with the differential methylation (PDF)

Accession ID of the RNA-seq and Methyl-seq data at the European Nucleotide Archive; raw nonmolecular data used in the study; and compounds of mixture and their nominal concentration, composition of the environmentally relevant pesticide mixture, list of differentially expressed genes, DEG with the F1:F0 treatment, differentially methylated regions and related genes, DMG with significant F1:F0 treatment interaction pattern of methylation, genes that were differentially methylated and expressed, and list of alternatively spliced genes (XLSX)

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Notes

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ABBREVIATIONS

DEG, differentially expressed gene
DMG, differentially methylated gene

DMR, differentially methylated region

MC, measured concentration

MCP larva, metamorphosis-competent pediveliger larva

NCSW, noncontaminated seawater

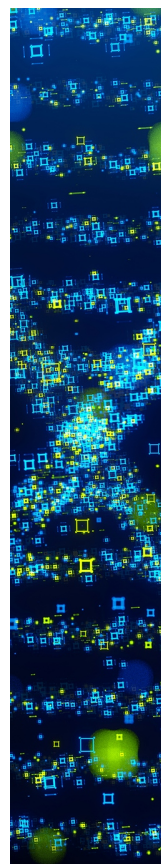
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